```
>> dynare mcmc.mod
Starting Dynare (version 6.4).
Calling Dynare with arguments: none
Starting preprocessing of the model file ...
Found 23 equation(s).
Evaluating expressions...
Computing static model derivatives (order 2).
Computing static model derivatives w.r.t. parameters (order 2).
Normalizing the static model...
Finding the optimal block decomposition of the static model...
5 block(s) found:
  3 recursive block(s) and 2 simultaneous block(s).
  the largest simultaneous block has 16 equation(s)
                                 and 16 feedback variable(s).
Computing dynamic model derivatives (order 2).
Computing dynamic model derivatives w.r.t. parameters (order 2).
Normalizing the dynamic model...
Finding the optimal block decomposition of the dynamic model...
5 block(s) found:
  3 recursive block(s) and 2 simultaneous block(s).
  the largest simultaneous block has 16 equation(s)
                                 and 16 feedback variable(s).
Preprocessing completed.
Preprocessing time: 0h00m00s.

STEADY-STATE RESULTS:

q                32.8507
k                4.41224
R                1.01351
b                143.013
x                0.0406367
varphi           0.0303425
mu               0.00686924
kp               2.20776
chi              1.02177
nu               0.00432944
eta              1.28813
zeta             1.02177
N                41.8193
phi              3.41978
Ne               41.5333
Nn               0.286025
xp               6.22046
bp               -101.193
Y                3.45318
eb               0
N_obs            0
Y_obs            0
q_obs            0

EIGENVALUES:
         Modulus             Real        Imaginary
               0                0                0
       3.882e-18       -3.882e-18                0
       1.364e-17        1.364e-17                0
       1.027e-14       -1.027e-14                0
       4.029e-09        5.157e-15        4.029e-09
```

```
      4.029e-09        5.157e-15      -4.029e-09
         0.1469           0.1469               0
         0.9804           0.9804               0
          1.018            1.018               0
           1.02             1.02               0
          1.022            1.022               0
          1.038            1.038               0
          3.365           -3.365               0
      1.193e+20       -1.193e+20               0
      2.084e+20       -2.084e+20               0

There are 7 eigenvalue(s) larger than 1 in modulus for 7 forward-looking variable(s)
The rank condition is verified.



======== Identification Analysis ========

Testing prior mean
-----------
The model does not solve for prior_mean (info = 4: Blanchard & Kahn conditions are not satisfied:
indeterminacy.)
-----------
Try sampling up to 50 parameter sets from the prior.
Found a random draw from the priors that solves the model:
    0.0066    0.0160    0.0078    5.6883    9.1694

Identification now continues for this draw.
Note that differences in the criteria could be due to numerical settings,
numerical errors or the method used to find problematic parameter sets.
Settings:
    Derivation mode for Jacobians:                         Analytic using sylvester equations
    Method to find problematic parameters:                 Nullspace and multicorrelation
coefficients
    Normalize Jacobians:                                   Yes
    Tolerance level for rank computations:                 robust
    Tolerance level for selecting nonzero columns:         1e-08
    Tolerance level for selecting nonzero singular values: 1e-03

REDUCED-FORM:
    All parameters are identified in the Jacobian of steady state and reduced-form solution
matrices (rank(Tau) is full with tol = robust).

MINIMAL SYSTEM (KOMUNJER AND NG, 2011):
    All parameters are identified in the Jacobian of steady state and minimal system
(rank(Deltabar) is full with tol = robust).

SPECTRUM (QU AND TKACHENKO, 2012):
    All parameters are identified in the Jacobian of mean and spectrum (rank(Gbar) is full with
tol = robust).

MOMENTS (ISKREV, 2010):
    All parameters are identified in the Jacobian of first two moments (rank(J) is full with tol
= robust).

==== Identification analysis completed ====
```

```
You did not declare endogenous variables after the estimation/calib_smoother command.
Initial value of the log posterior (or likelihood): -709.8975
-----------------
f at the beginning of new iteration,       709.8974914368
Predicted improvement:  1897518.423969072
lambda =          1; f =         998.3166625
lambda =    0.33333; f =         726.4148038
lambda =    0.11111; f =          61.5730733
lambda =   0.037037; f =         -81.5499767
lambda =   0.012346; f =        -196.9673555
lambda =  0.0041152; f =        -258.9235712
lambda =  0.0013717; f =        -226.0994543
lambda = 0.00045725; f =         -15.0125206
Norm of dx     19.481
----
Improvement on iteration 1 =      968.821062611
-----------------
f at the beginning of new iteration,      -258.9235711743
Predicted improvement:      219.855608610
lambda =          1; f =        -258.9071849
lambda =    0.33333; f =        -258.9224876
lambda =    0.11111; f =        -258.9235697
lambda =   0.037037; f =        -267.5810070
Norm of dx     0.2097
----
Improvement on iteration 2 =        8.657435859
-----------------
f at the beginning of new iteration,      -267.5810070331
Predicted improvement:      213.685236843
lambda =          1; f =        -267.5022638
lambda =    0.33333; f =        -267.5733843
lambda =    0.11111; f =        -267.5804838
lambda =   0.037037; f =        -267.5810051
lambda =   0.012346; f =        -271.4684799
lambda =   0.023866; f =        -260.3716489
lambda =    0.01607; f =        -271.1216840
Norm of dx    0.29134
----
Improvement on iteration 3 =        3.887472823
-----------------
f at the beginning of new iteration,      -271.4684798559
Predicted improvement:        0.800936602
lambda =          1; f =        -272.4140156
Norm of dx  0.0019807
----
Improvement on iteration 4 =        0.945535716
-----------------
f at the beginning of new iteration,      -272.4140155719
Predicted improvement:        1.005826090
lambda =          1; f =        -271.7112546
lambda =    0.33333; f =        -272.8153728
Norm of dx  0.0035188
----
Improvement on iteration 5 =        0.401357264
-----------------
f at the beginning of new iteration,      -272.8153728354
Predicted improvement:        0.030084836
```

```
lambda =          1; f =         -272.8605774
lambda =     1.9332; f =         -272.8764082
Norm of dx 0.00036962
----
Improvement on iteration 6 =        0.061035353
-----------------
f at the beginning of new iteration,      -272.8764081886
Predicted improvement:        0.029687333
lambda =          1; f =         -272.9343992
lambda =     1.9332; f =         -272.9859987
lambda =     3.7372; f =         -273.0790223
lambda =     7.2247; f =         -273.2346585
lambda =     13.967; f =         -273.4513050
Norm of dx 0.00056466
----
Improvement on iteration 7 =        0.574896822
-----------------
f at the beginning of new iteration,      -273.4513050109
Predicted improvement:        0.752532780
lambda =          1; f =         -274.7081543
lambda =     1.9332; f =         -275.3044281
Norm of dx    0.01745
----
Improvement on iteration 8 =        1.853123133
-----------------
f at the beginning of new iteration,      -275.3044281435
Predicted improvement:        0.510653465
lambda =          1; f =         -276.0440899
lambda =     1.9332; f =         -276.1042318
Norm of dx    0.01186
----
Improvement on iteration 9 =        0.799803694
-----------------
f at the beginning of new iteration,      -276.1042318377
Predicted improvement:        0.571725624
lambda =          1; f =         -276.3951914
lambda =    0.33333; f =         -276.3927697
lambda =    0.64439; f =         -276.4922792
Norm of dx   0.012688
----
Improvement on iteration 10 =        0.388047339
-----------------
f at the beginning of new iteration,      -276.4922791764
Predicted improvement:        0.022006277
lambda =          1; f =         -276.5100379
Norm of dx  0.0019886
----
Improvement on iteration 11 =        0.017758724
-----------------
f at the beginning of new iteration,      -276.5100378999
Predicted improvement:        0.002939570
lambda =          1; f =         -276.5141842
lambda =     1.9332; f =         -276.5149092
Norm of dx 0.00068191
----
Improvement on iteration 12 =        0.004871317
-----------------
f at the beginning of new iteration,      -276.5149092167
```

```
Predicted improvement:        0.002738632
lambda =          1; f =         -276.5202249
lambda =     1.9332; f =         -276.5248866
lambda =     3.7372; f =         -276.5330856
lambda =     7.2247; f =         -276.5459297
lambda =     13.967; f =         -276.5597421
Norm of dx   0.000918
----
Improvement on iteration 13 =        0.044832909
-----------------
f at the beginning of new iteration,      -276.5597421253
Predicted improvement:        0.046007055
lambda =          1; f =         -276.6503714
lambda =     1.9332; f =         -276.7324711
lambda =     3.7372; f =         -276.8846833
lambda =     7.2247; f =         -277.1562015
lambda =     13.967; f =         -277.6059866
lambda =         27; f =         -278.2488814
Norm of dx   0.020635
----
Improvement on iteration 14 =        1.689139237
-----------------
f at the beginning of new iteration,      -278.2488813623
Correct for low angle: 0.00330236
Predicted improvement:        3.233072521
lambda =          1; f =         -283.6603502
lambda =     1.9332; f =         -277.1515459
lambda =     1.3017; f =         -282.5642875
Norm of dx     1.1307
----
Improvement on iteration 15 =        5.411468831
-----------------
f at the beginning of new iteration,      -283.6603501928
Predicted improvement:        0.815064820
lambda =          1; f =         -283.4737479
lambda =    0.33333; f =         -284.0444766
lambda =    0.64439; f =         -284.0491722
Norm of dx    0.20642
----
Improvement on iteration 16 =        0.388822041
-----------------
f at the beginning of new iteration,      -284.0491722338
Predicted improvement:        0.646626354
lambda =          1; f =         -282.5272764
lambda =    0.33333; f =         -284.1464221
lambda =    0.11111; f =         -284.1539746
lambda =     0.2148; f =         -284.1850472
Norm of dx    0.42857
----
Improvement on iteration 17 =        0.135874954
-----------------
f at the beginning of new iteration,      -284.1850471882
Predicted improvement:        0.052642078
lambda =          1; f =         -284.2193841
Norm of dx   0.032283
----
Improvement on iteration 18 =        0.034336916
-----------------
```

```
f at the beginning of new iteration,      -284.2193841043
Predicted improvement:        0.005091417
lambda =          1; f =         -284.2249608
Norm of dx   0.002931
----
Improvement on iteration 19 =        0.005576647
-----------------
f at the beginning of new iteration,      -284.2249607513
Predicted improvement:        0.000238570
lambda =          1; f =         -284.2253292
lambda =     1.9332; f =         -284.2254877
Norm of dx 0.00093003
----
Improvement on iteration 20 =        0.000526982
-----------------
f at the beginning of new iteration,      -284.2254877333
Predicted improvement:        0.000449343
lambda =          1; f =         -284.2263338
lambda =     1.9332; f =         -284.2270354
lambda =     3.7372; f =         -284.2281494
lambda =     7.2247; f =         -284.2293885
Norm of dx  0.0007712
----
Improvement on iteration 21 =        0.003900770
-----------------
f at the beginning of new iteration,      -284.2293885036
Predicted improvement:        0.004635486
lambda =          1; f =         -284.2383483
lambda =     1.9332; f =         -284.2461613
lambda =     3.7372; f =         -284.2597653
lambda =     7.2247; f =         -284.2804594
lambda =     13.967; f =         -284.2995103
Norm of dx  0.0066968
----
Improvement on iteration 22 =        0.070121747
-----------------
f at the beginning of new iteration,      -284.2995102504
Correct for low angle: 0.00486669
Predicted improvement:        0.068402999
lambda =          1; f =         -284.4308028
lambda =     1.9332; f =         -284.5436174
lambda =     3.7372; f =         -284.7366710
lambda =     7.2247; f =         -285.0242890
lambda =     13.967; f =         -285.3049559
Norm of dx    0.08644
----
Improvement on iteration 23 =        1.005445686
-----------------
f at the beginning of new iteration,      -285.3049559362
Correct for low angle: 0.00416627
Predicted improvement:        0.274936939
lambda =          1; f =         -285.6817706
Norm of dx    0.46855
----
Improvement on iteration 24 =        0.376814616
-----------------
f at the beginning of new iteration,      -285.6817705521
Correct for low angle: 0.00336369
```

```
Predicted improvement:        0.272416262
lambda =          1; f =         -285.5798048
lambda =    0.33333; f =         -285.7807750
Norm of dx    0.69327
----
Improvement on iteration 25 =        0.099004423
-----------------
f at the beginning of new iteration,      -285.7807749751
Correct for low angle: 0.00265323
Predicted improvement:        0.255075800
lambda =          1; f =         -284.9357384
lambda =    0.33333; f =         -285.8239664
lambda =    0.11111; f =         -285.8240402
lambda =     0.2148; f =         -285.8390972
Norm of dx     0.5982
----
Improvement on iteration 26 =        0.058322233
-----------------
f at the beginning of new iteration,      -285.8390972084
Correct for low angle: 0.00307467
Predicted improvement:        0.139937904
lambda =          1; f =         -285.5965761
lambda =    0.33333; f =         -285.8655973
lambda =    0.11111; f =         -285.8624281
lambda =     0.2148; f =         -285.8707661
Norm of dx    0.52767
----
Improvement on iteration 27 =        0.031668888
-----------------
f at the beginning of new iteration,      -285.8707660960
Correct for low angle: 0.00271291
Predicted improvement:        0.109160706
lambda =          1; f =         -285.4262734
lambda =    0.33333; f =         -285.8780096
lambda =    0.11111; f =         -285.8879755
lambda =     0.2148; f =         -285.8909506
Norm of dx    0.45932
----
Improvement on iteration 28 =        0.020184496
-----------------
f at the beginning of new iteration,      -285.8909505916
Correct for low angle: 0.00205571
Predicted improvement:        0.108576102
lambda =          1; f =         -285.4904108
lambda =    0.33333; f =         -285.8828872
lambda =    0.11111; f =         -285.9056563
Norm of dx    0.40728
----
Improvement on iteration 29 =        0.014705747
-----------------
f at the beginning of new iteration,      -285.9056563384
Predicted improvement:        0.033159676
lambda =          1; f =         -285.9462298
Norm of dx    0.39342
----
Improvement on iteration 30 =        0.040573427
-----------------
f at the beginning of new iteration,      -285.9462297657
```

```
Predicted improvement:        0.005765382
lambda =          1; f =         -285.9529481
Norm of dx    0.18727
----
Improvement on iteration 31 =        0.006718361
-----------------
f at the beginning of new iteration,      -285.9529481268
Correct for low angle: 0.00173628
Predicted improvement:        0.000782945
lambda =          1; f =         -285.9474395
lambda =    0.33333; f =         -285.9526846
lambda =    0.11111; f =         -285.9530318
Norm of dx   0.038094
----
Improvement on iteration 32 =        0.000083685
-----------------
f at the beginning of new iteration,      -285.9530318117
Correct for low angle: 0.00298239
Predicted improvement:        0.000310632
lambda =          1; f =         -285.9517807
lambda =    0.33333; f =         -285.9530359
lambda =    0.11111; f =         -285.9530804
lambda =     0.2148; f =         -285.9530831
Norm of dx   0.033819
----
Improvement on iteration 33 =        0.000051280
-----------------
f at the beginning of new iteration,      -285.9530830918
Correct for low angle: 0.0013538
Predicted improvement:        0.000519984
lambda =          1; f =         -285.9494161
lambda =    0.33333; f =         -285.9529043
lambda =    0.11111; f =         -285.9531376
Norm of dx   0.026969
----
Improvement on iteration 34 =        0.000054491
-----------------
f at the beginning of new iteration,      -285.9531375825
Correct for low angle: 0.00380088
Predicted improvement:        0.000110196
lambda =          1; f =         -285.9529635
lambda =    0.33333; f =         -285.9531701
Norm of dx   0.024412
----
Improvement on iteration 35 =        0.000032478
-----------------
f at the beginning of new iteration,      -285.9531700605
Correct for low angle: 0.00118349
Predicted improvement:        0.000240816
lambda =          1; f =         -285.9516132
lambda =    0.33333; f =         -285.9531001
lambda =    0.11111; f =         -285.9531961
Norm of dx   0.016691
----
Improvement on iteration 36 =        0.000026023
-----------------
f at the beginning of new iteration,      -285.9531960835
Predicted improvement:        0.000028963
```

```
lambda =          1; f =         -285.9532266
Norm of dx   0.015089
----
Improvement on iteration 37 =        0.000030511
-----------------
f at the beginning of new iteration,      -285.9532265949
Predicted improvement:        0.000000065
lambda =          1; f =         -285.9532267
lambda =     1.9332; f =         -285.9532267
Norm of dx 0.00028579
----
Improvement on iteration 38 =        0.000000132
-----------------
f at the beginning of new iteration,      -285.9532267274
Correct for low angle: 0.00495877
Predicted improvement:        0.000000002
lambda =          1; f =         -285.9532267
lambda =    0.33333; f =         -285.9532267
lambda =    0.11111; f =         -285.9532267
lambda =   0.037037; f =         -285.9532267
lambda =   0.012346; f =         -285.9532267
Norm of dx 7.1403e-05
----
Improvement on iteration 39 =        0.000000000
improvement < crit termination

Final value of minus the log posterior (or likelihood):-285.953227

MODE CHECK

Fval obtained by the optimization routine: -285.953227

RESULTS FROM POSTERIOR ESTIMATION
parameters
         prior mean     mode    s.d.  prior pstdev

gamma_q      3.0000  -0.8686  1.9946   norm 5.0000
gamma_bY     0.0000   2.7949  0.8252   norm 5.0000

standard deviation of shocks
         prior mean     mode    s.d.  prior pstdev

eN           0.0100   0.0503  0.0051   invg 0.0050
eY           0.0100   0.0168  0.0016   invg 0.0050
eq           0.0100   0.0089  0.0013   invg 0.0050


Log data density [Laplace approximation] is 271.152930.

Estimation::mcmc: Multiple chains mode.
Estimation::mcmc: Old mh-files successfully erased!
Estimation::mcmc: Old metropolis.log file successfully erased!
Estimation::mcmc: Creation of a new metropolis.log file.
Estimation::mcmc: Searching for initial values...
Estimation::mcmc: Initial values found!

Estimation::mcmc: Write details about the MCMC... Ok!
Estimation::mcmc: Details about the MCMC are available in mcmc/metropolis\mcmc_mh_history_0.mat
```

```
Estimation::mcmc: Number of mh files: 1 per block.
Estimation::mcmc: Total number of generated files: 2.
Estimation::mcmc: Total number of iterations: 125000.
Estimation::mcmc: Current acceptance ratio per chain:
                                                       Chain  1: 46.2088%
                                                       Chain  2: 46.2184%
Estimation::mcmc: Total number of MH draws per chain: 125000.
Estimation::mcmc: Total number of generated MH files: 1.
Estimation::mcmc: I'll use mh-files 1 to 1.
Estimation::mcmc: In MH-file number 1 I'll start at line 25001.
Estimation::mcmc: Finally I keep 100000 draws per chain.



MCMC Inefficiency factors per block
Parameter       Block 1     Block 2
SE_eN            40.891      40.409
SE_eY            29.777      22.427
SE_eq            32.765      29.881
gamma_q          22.644      21.089
gamma_bY         33.001      29.150



Convergence diagnostics results for chain 1.

Geweke (1992) Convergence Tests, based on means of draws 25000 to 45000 vs 75000 to 125000 for
chain 1.
p-values are for Chi2-test for equality of means.
Parameter         Post. Mean        Post. Std   p-val No Taper   p-val 4% Taper   p-val 8% Taper
p-val 15% Taper
SE_eN                  0.053            0.006            0.076            0.743
0.740           0.735
SE_eY                  0.017            0.002            0.000            0.031
0.027           0.026
SE_eq                  0.009            0.001            0.000            0.104
0.100           0.074
gamma_q               -1.338            1.823            0.000            0.441
0.451           0.426
gamma_bY               3.142            0.835            0.000            0.136
0.135           0.112



Convergence diagnostics results for chain 2.

Geweke (1992) Convergence Tests, based on means of draws 25000 to 45000 vs 75000 to 125000 for
chain 2.
p-values are for Chi2-test for equality of means.
Parameter         Post. Mean        Post. Std   p-val No Taper   p-val 4% Taper   p-val 8% Taper
p-val 15% Taper
SE_eN                  0.053            0.006            0.000            0.261
0.223           0.190
SE_eY                  0.017            0.002            0.000            0.205
0.215           0.226
```

```
SE_eq                  0.009            0.001            0.007            0.592
0.583            0.564
gamma_q               -1.352            1.794            0.000            0.050
0.057            0.065
gamma_bY               3.147            0.817            0.000            0.027
0.032            0.036


Univariate convergence diagnostic, Brooks and Gelman (1998):
    Parameter 1...  Done!
    Parameter 2...  Done!
    Parameter 3...  Done!
    Parameter 4...  Done!
    Parameter 5...  Done!
警告: エクスポートされたイメージには座標軸ツール バーが表示されています。イメージから座標軸ツール バーを削
除するには、再度エクスポ
ートしてください。
marginal density: I'm computing the posterior mean and covariance...  Done!
marginal density: I'm computing the posterior log marginal density (modified harmonic mean)...
Done!


ESTIMATION RESULTS

Log data density (Modified Harmonic Mean) is 271.264004.
parameters
           prior mean   post. mean        90% HPD interval    prior       pstdev
gamma_q         3.000      -1.3599     -4.1675       1.6138     norm        5.0000
gamma_bY        0.000       3.1521      1.7915       4.4208     norm        5.0000

standard deviation of shocks
           prior mean   post. mean        90% HPD interval    prior       pstdev

eN              0.010       0.0532      0.0437       0.0625     invg        0.0050
eY              0.010       0.0172      0.0144       0.0198     invg        0.0050
eq              0.010       0.0092      0.0070       0.0113     invg        0.0050
警告: エクスポートされたイメージには座標軸ツール バーが表示されています。イメージから座標軸ツール バーを削
除するには、再度エクスポ
ートしてください。
警告: エクスポートされたイメージには座標軸ツール バーが表示されています。イメージから座標軸ツール バーを削
除するには、再度エクスポ
ートしてください。
警告: エクスポートされたイメージには座標軸ツール バーが表示されています。イメージから座標軸ツール バーを削
除するには、再度エクスポ
ートしてください。
--- Parameters Post. Std ---
gamma_q: 1.803086
gamma_bY: 0.824487
--- Shocks Post. Std ---
eN: 0.005921
eY: 0.001688
eq: 0.001326
Total computing time : 0h17m24s
Note: warning(s) encountered in MATLAB/Octave code
>>
```